Software Test Plan (STP) - Online Quiz System

Project: Online Quiz System
Version: 2.0
Authors: Debangana, Chinmay, Satvik, Sunil
Date: 06-11-2026
Status: Final for review

Version	Date	Change summary
1.0	28-09-2026	Baseline plan, TC-01 to TC-26
1.1	03-11-2026	Added TC-27 to TC-32 for the topic bank and randomised quiz
2.0	06-11-2026	Review update: template structure restored, all 32 test cases detailed, steps aligned with the implementation, results recorded from runs R1-R3

1. Introduction

Purpose: This document is the test plan for the Online Quiz System (IEEE 829 format). It describes what is tested, how, when and by whom, lists the 32 test cases and records their results.

Scope: Quiz flow, topic selection, randomised five-question attempts, answer validation, scoring, retry and navigation, administrator login and question management, and the non-functional and security requirements of SRS v2.0.

References: SRS v2.0; SAD v2.0; IEEE 829 Standard for Software Test Documentation; docs/test_execution_log.md and docs/test_summary_report.md in the project repository.

Definitions: MCQ (Multiple Choice Question), SRS, SAD, RTM (Requirements Traceability Matrix), TC (Test Case), Attempt (five questions on one topic), Fixture (temporary test question file).

2. Test Items

- QuizController (src/quiz_controller.cpp) - menus, topic selection, random selection, quiz flow, retry
- QuizLogic (src/quiz_logic.cpp) - answer checking and scoring
- QuestionLoader (src/question_loader.cpp) - reading and validating data/questions.txt
- AdminValidation (src/admin_validation.cpp) - question validation, save, modify, delete, credential checks
- Admin Module (src/main.cpp) - login with masked password and admin menu
- Question bank data/questions.txt (75 questions)

3. Features to be Tested

Features mapped to SRS requirement IDs (all 32 requirements are tested):

- Quiz Operation: OQ-F-001 to OQ-F-005, OQ-F-009, OQ-F-016, OQ-F-022
- Answer Checking and Scoring: OQ-F-006 to OQ-F-008, OQ-F-010, OQ-F-011

- Topic Bank and Randomisation: OQ-F-017 to OQ-F-021
- Question Management: OQ-F-012 to OQ-F-014
- User and Access Control: OQ-F-015
- OQ-NF-001 Response time; OQ-NF-002 Score reliability; OQ-NF-003 Clear messages; OQ-NF-004 No exposure of admin data; OQ-NF-005 Portability
- Security: OQ-SR-001 to OQ-SR-005 (section 5.1)

4. Features Not to be Tested

- Web, GUI and network features - not part of the system.
- Multi-user or simultaneous access - single-user console program.
- Compiler, standard library and operating-system internals - third-party.
- Strength of the hard-coded test credentials - a known limitation (SRS 5.1.3), not a test target.

5. Test Approach / Strategy

Levels:

- Unit tests - tests/test_main.cpp, test_admin_validation.cpp, test_question_loader.cpp and test_score_reliability.cpp, each compiled with all source files except main.cpp.
- Integration tests - QuizController with QuestionLoader and QuizLogic through injected input/output streams.
- System tests - tests/system/run_system_tests.sh builds the program and runs the scripted cases against the real binary, using a temporary copy of the question file so data/questions.txt is never changed.
- Acceptance tests - demo to the instructor using the suites in SRS section 6.

Types:

- Functional testing of every OQ-F requirement.
- Negative and boundary testing - invalid answers (E, 5, @, AB, empty), invalid topics (9, abc), malformed records, a topic with only 4 questions, wrong or empty credentials, hidden menu numbers.
- Randomisation testing - check invariants (5 distinct questions, all from the chosen topic, new set on retry) over repeated runs instead of a fixed order.
- Regression testing - the full script is re-run after every fix (run R2 after the TC-12 fix).
- Performance, reliability, usability and portability testing for the NFRs; security testing (5.1).

Entry Criteria: Program builds with -std=c++17 -Wall -Wextra; test fixtures and scripts available; test cases reviewed.

Exit Criteria: All 32 test cases executed and recorded; all High-priority and security cases passed; no open Critical or High defect; RTM updated with results.

5.1 Security Validation

Each security requirement (SRS 5.1.2) is checked against its objective. All security test cases must pass before release.

- OQ-SR-001 (SO-1, SO-3) Login before management - choose Admin login; the login prompt must appear before any admin menu (TC-22).

- OQ-SR-002 (SO-1) Reject wrong credentials - wrong password, wrong username and empty fields must be refused (TC-23).
- OQ-SR-003 (SO-1) Validate admin input - empty text, a missing option and answer E must be rejected and the file left unchanged (TC-24).
- OQ-SR-004 (SO-3) No direct admin access - admin, 9, 99 and -1 at the main menu, and menu numbers at an answer prompt, must not open admin functions (TC-25).
- OQ-SR-005 (SO-2) Hide credentials - the password must never appear in output (automated) and must show as "*" in a real terminal (manual) (TC-26).
- Related checks: no route to management without login (OQ-F-015, TC-15); no answers or admin options in the user flow (OQ-NF-004, TC-20).

6. Test Environment

Hardware: Standard laptop with keyboard and display.

Software: macOS (Darwin 25.6) with Apple clang 21.0; Ubuntu 22.04 with g++ 11.4; Windows with MinGW g++ (manual run). Build flags -std=c++17 -Wall -Wextra.

Tools: Bash system-test script with input redirection, C++ assert-based unit tests, GitHub Issues for defects.

Test Data: Delivered bank (15 questions x 5 topics); temporary fixtures of 4-6 ML questions with every correct answer B (needed because question order is random); fixture with malformed records; tests/data/answers_3of5.txt; test account admin / admin123.

7. Test Schedule

Milestones:

- Test case design (TC-01 to TC-26): 28-09-2026
- Sprint 1 release v1.0: 24-10-2026
- Scripted execution (R1) and regression run (R2): 30-10-2026
- Test cases for the topic feature (TC-27 to TC-32): 03-11-2026
- Full run on Linux including TC-27 to TC-32 (R3): 06-11-2026
- Manual checks TC-19, TC-21 (Windows) and TC-26 (masking): before the final demo

8. Test Deliverables

- Test Plan (this document)
- Test Cases (section 13.2)
- Test Scripts (tests/system/run_system_tests.sh, unit tests)
- Test Data (fixtures, tests/data/answers_3of5.txt)
- Test Execution Log (docs/test_execution_log.md)
- Defect Reports (GitHub Issues)
- Test Summary Report (docs/test_summary_report.md)

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	<Name>	Prepare plan, coordinate execution, update RTM
Test Engineer	<Name>	Write scripts, execute test cases, log defects

Role	Name	Responsibility
Developer	<Name>	Fix defects, maintain unit tests
Product Owner	<Name>	Approve results, sign off readiness

10. Risks and Mitigation

Risk	Mitigation
Random question order makes expected output unpredictable	Fixtures with one answer key (B); check invariants, not a fixed order
Tests change the real question bank	System script works on a temporary copy of the file
Password masking cannot be checked with redirected input	Automated check that the password is never printed, plus a manual terminal check
Team hardware is mostly macOS	Linux run R3; Windows MinGW run done by hand

11. Assumptions & Dependencies

- A C++17 compiler is installed on each test machine.
- Tests are run from the repository root.
- Requirement IDs in SRS v2.0 do not change during testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The build fails (the system script stops automatically).
- A defect blocks more than 30% of the test cases.

Resume testing if:

- The blocking defect is fixed and the build passes.
- The full script is re-run as a regression run.

13. Test Case Management & Traceability

Test runs used for the results in this section:

Run	Date	Environment	Build	Scope	Outcome
R1	30-10-2026	macOS (Darwin 25.6), Apple clang 21.0	main before PR #50	TC-01 to TC-26	22 pass, 1 fail (TC-12), 3 manual/partial
R2	30-10-2026	macOS (Darwin 25.6), Apple clang 21.0	main after PR #50 (regression)	TC-01 to TC-26	23 pass, 0 fail, 3 manual/partial
R3	06-11-2026	Ubuntu 22.04, g++ 11.4	commit 05bc7d4	TC-01 to TC-32	29 pass, 0 fail, 3 manual/partial

13.1 Requirements Traceability Matrix

Req ID	Requirement	Module	Test case	Result
OQ-F-001	Start quiz	QuizController	TC-01	Pass
OQ-F-002	One question at a time	QuizController	TC-02	Pass
OQ-F-003	Four options A-D	QuizController	TC-03	Pass
OQ-F-004	Accept answer	QuizController	TC-04	Pass

Req ID	Requirement	Module	Test case	Result
OQ-F-005	Reject invalid option	QuizController	TC-05	Pass
OQ-F-006	Check answer	QuizController, QuizLogic	TC-06	Pass
OQ-F-007	Score +1 if correct	QuizLogic	TC-07	Pass
OQ-F-008	Score same if wrong	QuizLogic	TC-08	Pass
OQ-F-009	Next question	QuizController	TC-09	Pass
OQ-F-010	Final score	QuizLogic	TC-10	Pass
OQ-F-011	Show final score	QuizController	TC-11	Pass
OQ-F-012	Admin add question	Admin Module (main.cpp), AdminValidation	TC-12	Pass
OQ-F-013	Admin modify question	Admin Module (main.cpp), AdminValidation	TC-13	Pass
OQ-F-014	Admin delete question	Admin Module (main.cpp)	TC-14	Pass
OQ-F-015	Block unauthenticated users	QuizController, Admin Module (main.cpp)	TC-15	Pass
OQ-F-016	Reset score on new attempt	QuizController, QuizLogic	TC-16	Pass
OQ-F-017	Topic question bank	QuestionLoader, Question bank	TC-27	Pass
OQ-F-018	Question record format	QuestionLoader, AdminValidation	TC-28	Pass
OQ-F-019	Topic selection	QuizController, QuestionLoader	TC-29	Pass
OQ-F-020	Five random questions	QuizController	TC-30	Pass
OQ-F-021	Avoid repeating previous set	QuizController	TC-31	Pass
OQ-F-022	Result menu	QuizController	TC-32	Pass
OQ-NF-001	Response <= 1 s	QuizController	TC-17	Pass
OQ-NF-002	Score reliability	QuizLogic	TC-18	Pass
OQ-NF-003	Clear messages	QuizController, Admin Module (main.cpp)	TC-19	Pending
OQ-NF-004	No exposure of admin data	QuizController	TC-20	Pass
OQ-NF-005	Portability	All modules	TC-21	Partial
OQ-SR-001	Login before management	Admin Module (main.cpp)	TC-22	Pass
OQ-SR-002	Reject wrong credentials	Admin Module (main.cpp), AdminValidation	TC-23	Pass
OQ-SR-003	Validate admin input	Admin Module (main.cpp), AdminValidation	TC-24	Pass
OQ-SR-004	No direct admin access	QuizController	TC-25	Pass
OQ-SR-005	Hide credentials	Admin Module (main.cpp)	TC-26	Partial

Coverage: 32 / 32 requirements mapped to a test case (100%). **Results:** 29 passed, 0 failed, 3 partly or not yet verified (TC-19, TC-21, TC-26 - manual checks).

13.2 Test Cases

32 test cases: 22 functional, 5 non-functional and 5 security.

Test case	Req.	Preconditions and steps	Expected result	Result
TC-01 Start a new quiz Functional, High	OQ-F-001	**Pre:** Fixture bank: ML questions, every correct answer B. 1. At the main menu enter 2. 2. Enter 1 (ML).	Topic menu, then "Question 1 of 5" with score 0.	Pass (R2, R3)

Test case	Req.	Preconditions and steps	Expected result	Result
TC-02 One question at a time Functional, High	OQ-F-002	**Pre:** Quiz started (TC-01). 1. Observe the screen. 2. Answer Question 1.	Only Question 1 shown; Question 2 appears after answering.	Pass (R2, R3)
TC-03 Four options shown Functional, High	OQ-F-003	**Pre:** Quiz started. 1. Check each of the 5 questions.	Each question shows exactly options A, B, C, D.	Pass (R2, R3)
TC-04 Valid answer accepted Functional, High	OQ-F-004	**Pre:** Quiz at Question 1. 1. Enter B.	B is accepted and processed.	Pass (R2, R3)
TC-05 Invalid option rejected Negative / Boundary, High	OQ-F-005	**Pre:** Quiz at Question 1. 1. Enter E, 5, @, AB and an empty line. 2. Enter A.	Each invalid input shows "Invalid option. Enter A, B, C, or D."; Question 1 stays; A is accepted.	Pass (R2, R3)
TC-06 Answer compared with key Functional, High	OQ-F-006	**Pre:** Fixture bank: ML questions, every correct answer B. 1. Answer Q1 with B. 2. Answer Q2 with A.	Q1 "Correct!", Q2 "Incorrect!"; correct answer shown after each.	Pass (R2, R3)
TC-07 Score +1 on correct answer Functional, High	OQ-F-007	**Pre:** Fixture bank: ML questions, every correct answer B. 1. Answer all questions B.	Score increases by exactly one per correct answer (5 / 5).	Pass (R2, R3)
TC-08 Score unchanged on wrong answer Functional, High	OQ-F-008	**Pre:** Fixture bank: ML questions, every correct answer B. 1. Answer four questions B and one A.	Final score 4 / 5.	Pass (R2, R3)
TC-09 Move to next question Functional, High	OQ-F-009	**Pre:** Quiz at Q1. 1. Enter E. 2. Enter A.	Stays on Q1 after E; Q2 shown after A.	Pass (R2, R3)
TC-10 Final score calculated Functional, High	OQ-F-010	**Pre:** tests/data/answers_3of5.txt. 1. Complete the quiz with the script.	Final score 3.	Pass (R2, R3)
TC-11 Final score displayed Functional, High	OQ-F-011	**Pre:** Quiz completed with 3 correct. 1. Read the result screen.	"Your score: 3 / 5".	Pass (R2, R3)
TC-12 Admin adds a question Functional, High	OQ-F-012	**Pre:** Logged in as admin; 5 questions in bank. 1. Choose 1 (Add). 2. Enter topic ML, text, options and answer B. 3. Start an ML quiz.	Saved with topic (6 questions) and used in the quiz.	Pass (failed in R1, fixed in PR #50) (R1, R2, R3)
TC-13 Admin modifies a question Functional, High	OQ-F-013	**Pre:** Logged in as admin. 1. Choose 2 (Modify), select Q2. 2. Enter new text and answer. 3. Start a quiz.	Updated text and answer saved and used.	Pass (R2, R3)
TC-14 Admin deletes a question Functional, High	OQ-F-014	**Pre:** Logged in; 6 questions. 1. Choose 3 (Delete), select Q6. 2. Confirm with Y.	Q6 removed; 5 remain.	Pass (R2, R3)
TC-15 Unauthenticated user blocked Negative / Security, High	OQ-F-015	**Pre:** Main menu, not logged in. 1. Try to reach add, modify and delete without logging in.	No route exists; other input gives "Invalid menu choice."; data unchanged.	Pass (R2, R3)
TC-16 Score reset on retry Functional, Medium	OQ-F-016	**Pre:** Attempt completed. 1. Choose 1 (Retry this topic).	New attempt starts at Q1 with score 0.	Pass (R2, R3)

Test case	Req.	Preconditions and steps	Expected result	Result
TC-17 Response time Performance, Medium	OQ-NF-001	**Pre:** 20 scripted inputs. 1. Measure total processing time in the test script.	>= 95% of inputs within 1 s.	Pass (20 inputs: 14 ms in R2, 5 ms in R3) (R2, R3)
TC-18 Score reliability Reliability, High	OQ-NF-002	**Pre:** answers_3of5.txt. 1. Run 3 times with the same answers.	Score 3 on every run.	Pass (R2, R3)
TC-19 Clear messages Usability (manual), Medium	OQ-NF-003	**Pre:** 3 students new to the program. 1. Each completes a quiz without help.	All complete it without extra instructions.	Pending (manual) (-)
TC-20 Admin data hidden from user flow Security, High	OQ-NF-004	**Pre:** Normal user. 1. Go through all user menus and a quiz.	No correct answer before answering; no management option visible.	Pass (R2, R3)
TC-21 Builds on standard machines Portability, Medium	OQ-NF-005	**Pre:** macOS (clang++), Linux (g++), Windows (MinGW). 1. Build with -std=c++17 -Wall -Wextra. 2. Run one complete quiz.	0 errors, 0 warnings; quiz runs on all three.	Partial: macOS pass (R2), Linux pass (R3), Windows pending (R2, R3)
TC-22 Login required Security, High	OQ-SR-001	**Pre:** Not logged in. 1. Choose 1 (Admin login).	Login prompt first; admin menu only after correct login.	Pass (R2, R3)
TC-23 Wrong credentials rejected Security / Negative, High	OQ-SR-002	**Pre:** Test account admin / admin123. 1. Wrong password. 2. Wrong username. 3. Empty fields. 4. Correct credentials.	First three rejected with the messages in OQ-SR-002; correct login opens the admin menu.	Pass (R2, R3)
TC-24 Admin input validated Security / Negative, Medium	OQ-SR-003	**Pre:** Logged in. 1. Add with empty text. 2. Add with a missing option. 3. Add with answer E.	Each rejected; questions.txt unchanged.	Pass (R2, R3)
TC-25 No direct admin access Security / Negative, High	OQ-SR-004	**Pre:** Normal user. 1. At the main menu enter admin, 9, 99, -1. 2. During a quiz enter 1 at an answer prompt.	All rejected; admin menu never opens.	Pass (R2, R3)
TC-26 Credentials hidden Security, Medium	OQ-SR-005	**Pre:** Real terminal (not redirected input). 1. Type the password at the login prompt. 2. Check all output.	Password shown as "*"; never printed.	Partial: output check pass (R2, R3); "*" masking to be confirmed by hand (R2, R3)
TC-27 Topic bank contents Data, High	OQ-F-017	**Pre:** Delivered data/questions.txt. 1. Count records per topic.	15 each for ML, DBMS, SE, DSA, GK (75).	Pass (15 x 5 = 75) (R3)
TC-28 Record validation Negative / Data, High	OQ-F-018	**Pre:** 5 valid ML records + 4 malformed (6 fields, topic XYZ, answer E, empty text). 1. Start an ML quiz.	4 warnings with line numbers; only the 5 valid questions used.	Pass (R3)
TC-29 Topic selection and filtering Functional / Negative, High	OQ-F-019	**Pre:** Delivered bank. 1. Take one attempt for each topic 1-5. 2. Enter 9 and abc at the topic menu.	All questions belong to the chosen topic; invalid choices give "Invalid topic selection.".	Pass (R3)

Test case	Req.	Preconditions and steps	Expected result	Result
TC-30 Five distinct random questions Functional / Boundary, High	OQ-F-020	**Pre:** Delivered bank; then a bank with only 4 ML questions. 1. Run 20 attempts. 2. Start ML with 4 questions.	5 distinct questions in all 20; with 4 questions "Not enough valid questions for topic ML..." and no attempt.	Pass (R3)
TC-31 Retry gives a new set Functional, Medium	OQ-F-021	**Pre:** Delivered bank (15 ML). 1. Complete an ML attempt and retry; repeat 10 times.	Consecutive sets differ in all 10; score reset; bank unchanged.	Pass (R3)
TC-32 Result menu navigation Functional, High	OQ-F-022	**Pre:** Attempt completed. 1. Choose 1, then 2, then 0.	Retry starts a new attempt; 2 shows the topic menu; 0 returns to the main menu.	Pass (R3)

14. Test Metrics & Reporting

Metrics:

- Test cases executed: 32 / 32 (TC-19 manual check still to be recorded)
- Passed: 29; failed: 0; partial or pending: 3
- Defects: 1 Medium (fixed), 1 Low (open); no Critical or High defect
- Requirement coverage: 32 / 32 (100%)

Defect	Found by	Severity	Status
Admin-added questions were saved without their topic, so they never appeared in a quiz	TC-12 (R1)	Medium	Fixed in PR #50; verified in R2 and R3
tests/test_question_loader.cpp still uses the old 6-field record format, so its assertion fails	Unit test run	Low	Open (test code only; loader behaviour verified by TC-28)

Reports:

- Execution log after each run (docs/test_execution_log.md)
- Test Summary Report (docs/test_summary_report.md)

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		